The human ear

1 Use the following words to complete the passages that follow.

AUDITORY	BRAIN	CELLS	COCHLEA	EUSTACHIAN	HAIRS	HEAR
HERTZ	HIGH	MAGNIFY	MEMBRANE	NERVE	NEURAL	OSSICLES
OVAL	PINNA	SOUND	THREE	ULTRASONIC	BASILAR	

The human ear can be divided into _____ parts: the outer ear, middle ear and inner ear. The _____ (auricle) is an outer flap of cartilage which collects sound waves and passes them into the _____ canal. At the end of this canal (approximately 3 cm in length) is the tympanic _____ or eardrum. When _____ waves strike the taut skin of the eardrum they cause it to vibrate. The middle ear has three tiny bones (_____) which vibrate when the eardrum does. The ossicles act as levers to _____ movement. The three tiny bones are the hammer (*malleus*), anvil (*incus*) and stirrup (*stapes*), named after the objects they resemble in shape. The amplified signal is conducted to the _____ window (*fenestra ovalis*) that connects the middle and inner ears. The _____ tube equals the pressure on both sides of the eardrum.

An important part of the inner ear is the _____, which resembles a snail shell and is filled with fluid. Alfonso Giacomo Gaspare Corti was an Italian anatomist who discovered (in 1850) the complex organ of the ear by which sound waves are converted to _____ impulses. The organ of Corti contains fine _____ that vibrate in response to sound waves. These hair _____ (located on the basilar membrane) vibrate according to their length. Short hairs vibrate in response to _____ frequencies and are located nearest the oval window. Those longer hairs at the end of the _____ membrane respond to lower frequency sounds. Each hair cell sends a signal to the cochlear (auditory) _____ which transmits the information to the _____ .

Humans can only _____ a small range of sounds. The range of human hearing tends to be from 20 _____ to 20 kilohertz. Frequencies above 20 kHz are _____.

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Complete the following table that shows the names, location and functions of parts of the human ear. The first row has already been done for you.

Part	Location	Function
External ear (*pinna*)	Outer ear	Directs sounds to auditory canal.
2	Outer ear	Takes sound to the eardrum.
Eardrum (*tympanic membrane*)	**3**	Vibrates when sound hits it.
Eustachian tube	Middle ear	**4**
5 (*malleus*)	Middle ear	Lever that magnifies movement due to sound.
Anvil (*incus*)	**6**	Lever that magnifies movement due to sound.
Stirrup (*stapes*)	Middle ear	**7**
8 (*fenestra ovalis*)	Inner ear	Connects to the throat using the Eustachian tube.
Cochlea	**9**	Turns vibrations to electrical signals.
Semicircular canals	Inner ear	**10**
11	Inner ear	Carries nervous signal from cochlea to the brain.

Decide whether the following statements are true or false:

12 Sound waves have kinetic energy. _____

13 The cochlea is filled with air. _____

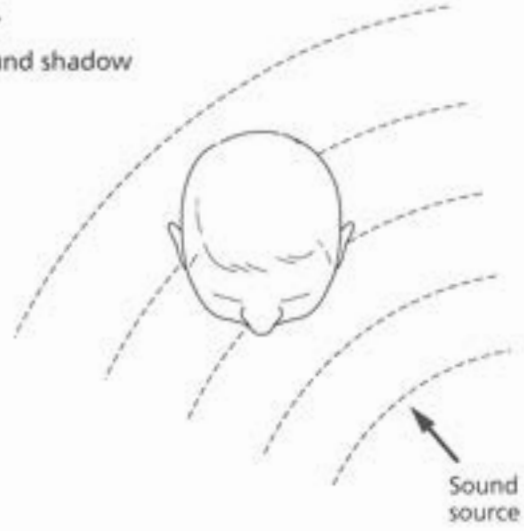
14 The ossicles are made of cartilage. _____

15 The term *cochlea* comes from the Greek word meaning snail. _____

16 The organ of Corti is located in the semicircular canals. _____

The diagram shows a phenomenon called 'sound shadow'.

17 Referring to the diagram explain what is meant by sound shadow and the role it plays in the hearing of mammals.



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